LOGO

2024

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Airline Contrail Impact Report 2024

What are Contrails?

Contrails — the thin, white lines you sometimes see behind airplanes — have a surprisingly large impact on our climate. Contrails warm the planet because contrail clouds act like a blanket on Earth and have a net heating effect. The 2022 IPCC report noted that clouds created by contrails account for roughly 35% of aviation's global warming impact — over half the impact of the world's jet fuel. Find more info about contrails and the climate on our website contrails.org.

What is Global Warming Potential (GWP)?

GWP measures how much warming contrails cause over a number of years compared to CO2. Contrails heat the Earth quickly but for a short time, and GWP helps compare their short-term impact to the longer-lasting greenhouse gas, CO2.

In this report we initially show the contrail impact in CO2e over 20, 50 and 100 years to align with the guidelines from the EU Non-CO2 MRV report starting in 2025. Wherever we only show one value for CO2e we use the middle value, GWP50, as default.

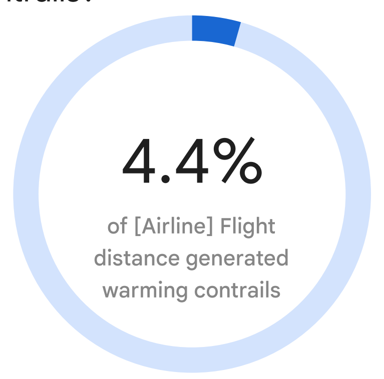
Impact Data

2024

Based on our prediction model, 5.5 Million km (55,501 flight hours) or 4.4% of all [Airline] flights generated warming contrails in 2024.

# of Flights	Flight hours	Contrails (GWP 50)	Fuel burn
15,380 flights	55,010 hours	275k metric tons CO2e	954k metric tons CO2

What percentage of [Ariline] flights created warming contrails?



Distance (km) with warming contrails

Distance (km) without warming contrails

How many flight kilometers of warming contrails?

Flight kilometers for all flights

46M km

74.3% | 25.7%

Flight kilometers creating warming contrails

5.5M km

84.7% | 15.3%

Nightime · Daytime

Impact Data: intra-European flights only

2024

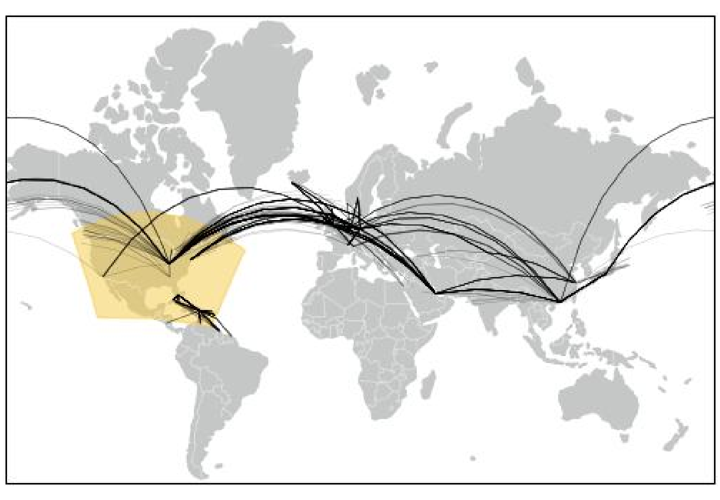
Based on our prediction model, this is the impact from the DHL flights that are included in the EU's non-CO2 reporting requirements. The EU ETS area covers flights within and between countries in the European Economic Area (EEA, which consists of EU member states and Iceland, Norway, and Liechtenstein), and from the EEA to the UK and Switzerland. It also covers the EU's nine, socalled outermost regions: French Guiana, Guadeloupe, Martinique, Mayotte, Reunion Island, Saint-Martin, Azores, Madeira, and The Canary Islands.

# of Flights	Flight hours
7,880 flights	25,010 hours

Contrails (GWP 50)	Fuel burn
135k metric tons CO2e	454k metric tons CO2

Observation coverage area & verification

The yellow area shows the coverage region where our satellite imagery based verification has been validated. For the rest of the world, we use our algorithm predictions.



Observations area · Algorithm prediction area · Flight paths

Predicted contrails in observation area	Verified contrails kilometers
1,218k km	1,850k km

Contrail warming using different time horizons: GWP20, GWP50, and GWP100.

There is no single "correct" way to convert contrail warming to CO2e. This is partly because the lifetime of a single contrail (hours) is much shorter than the lifetime of CO2 in the atmosphere (hundreds to thousands of years). So when using the Global Warming Potential (GWP) metric and comparing contrail warming to the warming from CO2 over 20 years, the contrail warming will be about four times higher than if comparing to CO2 over 100 years. We use GWP20, GWP50, and GWP100 to align with the EU MRV guidelines. The middle value, GWP50, is used as the default in the report.

GWP 100

157k metric tons

GWP 50

275k metric tons

GWP 20

578k metric tons

Fuel emissions (CO2) vs contrail warming (CO2e)

GWP 50

The contrail warming impact is often lower in the summer time and higher in dark months. This is because contrail clouds that persist in the dark are the most warming.

Fuel Emissions	Contrails
620,493 t CO2	60,314 t CO2e
77.6%	22.4%

Contrail warming - daytime vs nighttime

GWP 50

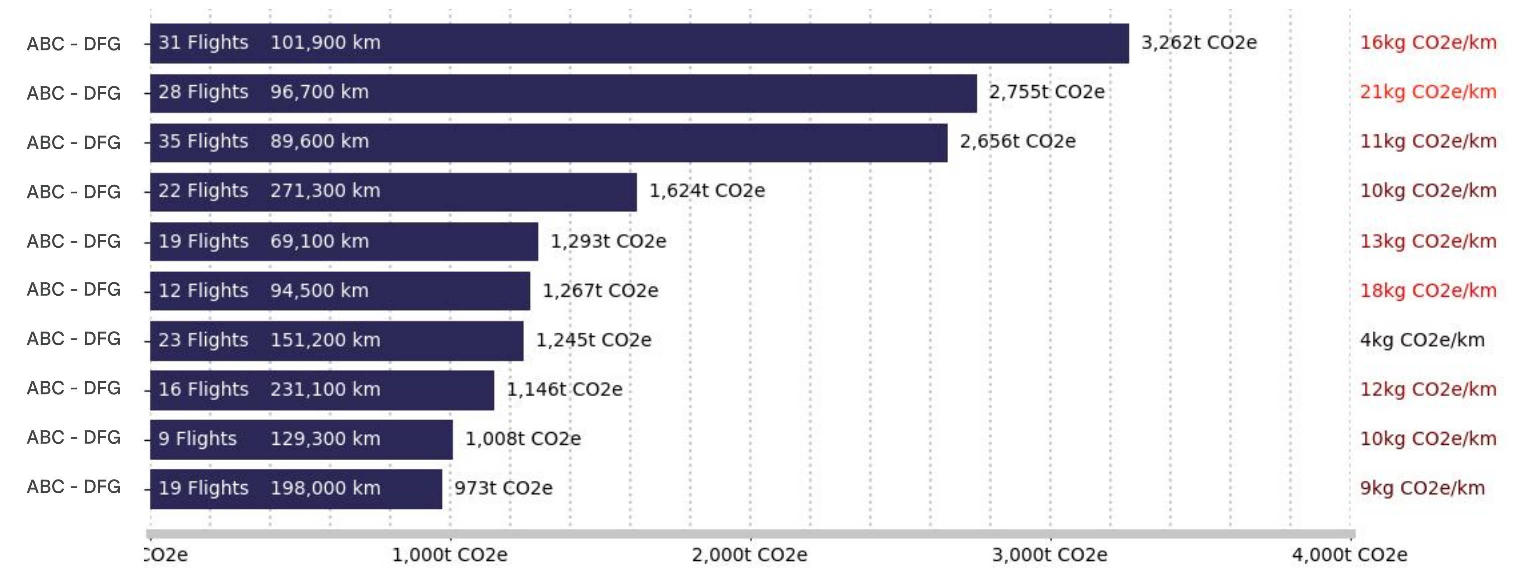
In the daytime, contrails sometimes have a cooling effect when reflecting some of the sun's heat back into space. But at all times, contrails have a warming effect by acting like a blanket on Earth. This is evident at night when there is no sunlight to reflect, and all contrails are warming.

Nighttime	Daytime
260,082 t CO2e	10,650 t CO2e
94%	6%

Origin-Destination pairs with the highest average total contrail warming (CO2e)

2024 · GWP 50

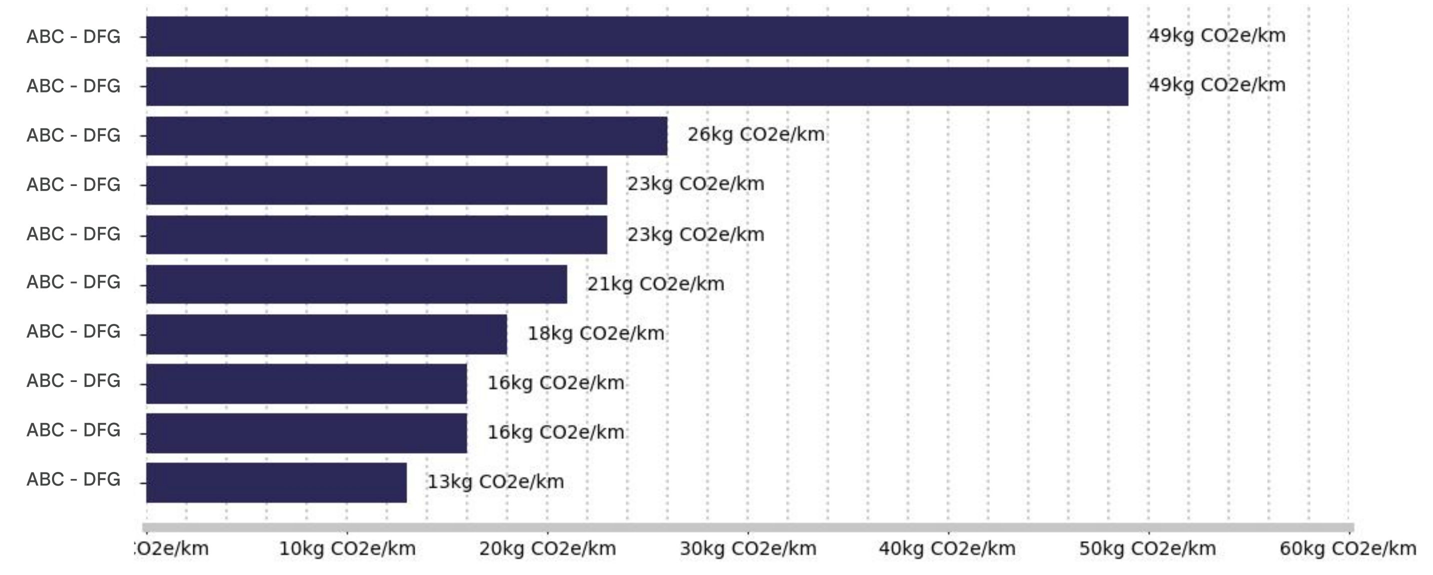
These ten OD pairs are responsible for 63% of [Airline]'s total contrail warming.
The most warming OD pairs are often very long flights where the majority of the journey takes place in the dark, when contrail are most warming



Origin-Destination pairs with the highest average total contrail warming per flown kilometer (CO2e/km)

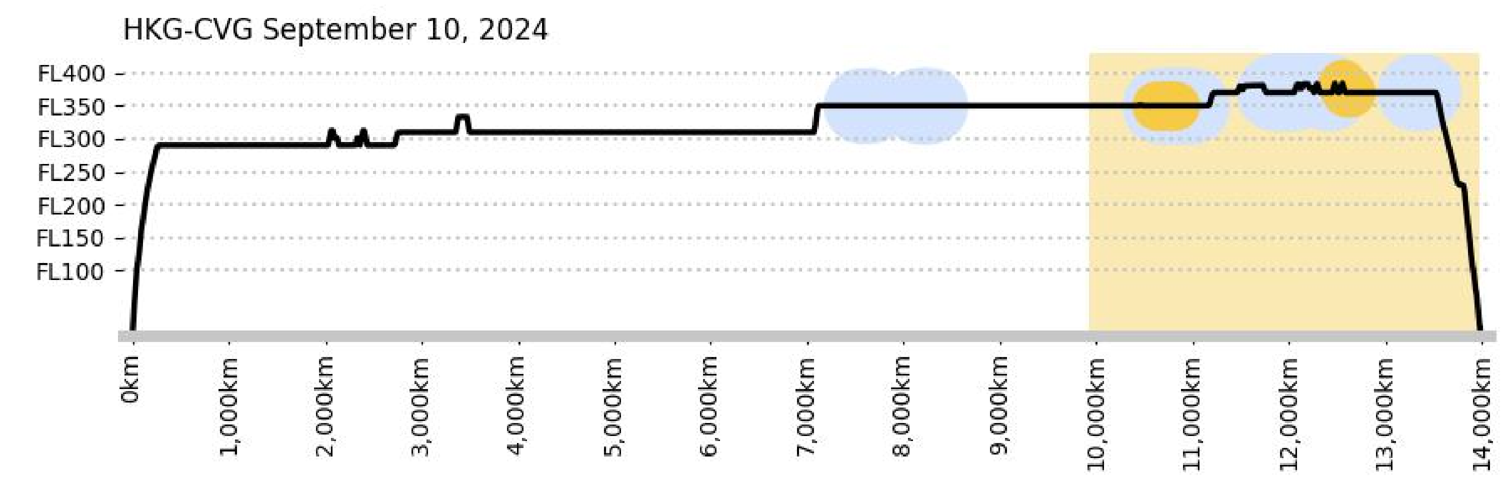
2024 · GWP 50

The average carbon dioxide emissions per kilometer for [Airline] in September was 21 kg CO2/km. The OD pair with the highest contrail warming per kilometer is EMA - CPH adding 49 kg CO2e/km - or 2.3 times the average warming from the CO2 alone.
The most warming OD pairs per flown kilometer are often flights that fly through contrail-prone zones (for example the Eastern part of the US) at night, when contrails are most warming.



Case study: predicted vs verified contrails

The light yellow color shows the observation area, and dark yellow indicates verified contrail formation from satellite imagery within the observation area.
The light blue areas indicate where contrails are predicted to appear.



Flight path · Predicted contrails · Confirmed contrails · Observation region

Did you know?

Some flight planning software providers, like Flightkeys and CAE, have implemented contrail avoidance in their flight planning tools (or are about to).

In 2023, American Airlines, Google Research, and Reviate conducted a trial in which they avoided 54% of contrail kilometers by flying under contrail-prone areas.

In 2024, an extensive study of over 84,000 flights showed that, theoretically, it was possible to eliminate 73% of the contrail warming from these flights by spending 0.11% more jet fuel to adjust some of the flight paths.

See where contrails are forming right now on this world map of contrails.

Read more about contrails on our websites: Reviate, Google Research.